

Alzheimer's disease

Alzheimer's disease is a progressive neurodegenerative disorder and the most common cause of dementia. It slowly destroys memory, thinking skills, and the ability to perform basic daily tasks. While there is no cure, early detection and management can greatly improve quality of life and help patients plan for the future.

Symptoms & Progression

The disease worsens over time, starting with mild forgetfulness and advancing to a loss of independence. Symptoms typically fall into three main areas:

- **Memory Loss:** Difficulty recalling recent conversations or events, repeatedly asking the same questions, and frequently misplacing items.
- **Cognitive Decline:** Trouble concentrating, managing finances, and expressing thoughts or finding the right words.
- **Behavioural Changes:** Unexplained mood swings, confusion about time or place, and changes in personality.

What is a stroke?

A stroke is a medical emergency that happens when something prevents your brain from getting enough [blood](#) flow. A blocked [blood vessel](#) or bleeding in your [brain](#) can cause strokes.

Healthcare providers sometimes refer to strokes as cerebrovascular accidents (CVAs) or brain attacks.

Strokes are the second leading cause of death worldwide and the fifth most common in the U.S.

If you think you or someone you're with is having a stroke, immediately call 911 or your local emergency services number. Strokes are life-threatening and can be fatal. The sooner someone is diagnosed and treated, the more likely it is they'll survive a stroke. Every second counts.

Symptoms and Causes

Aphasia (trouble speaking or a complete loss of speech)

Blurry vision or double vision (diplopia)

- Confusion or agitation
- Coma
- Dizziness or vertigo
- Headaches (usually sudden and severe)
- Loss of muscle control on one side of your face
- Loss of coordination or clumsiness (ataxia)
- Memory loss (amnesia)

- Mood swings or sudden personality changes
- Nausea and vomiting
- Neck stiffness
- Passing out or fainting
- Seizures
- Slurred or garbled speaking (dysarthria)
- Sudden worsening or loss of your senses (including vision, hearing, smell, taste and touch)
- Weakness or paralysis on one side of your face and body

Bell's palsy is a temporary, sudden weakness or paralysis of facial muscles caused by inflammation of the 7th cranial nerve, often linked to viral infections. Symptoms include a drooping eyelid/mouth, drooling, and taste changes, usually appearing within 48-72 hours. Most recover fully within 3 weeks to 6 months, though some cases may take up to a year.

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Key Details About Bell's Palsy

- **Causes:** The exact cause is often unknown, but it is believed to be triggered by viral infections (like herpes simplex) that cause the facial nerve to swell and become compressed within its bony canal.
- **Symptoms:**
 - Sudden weakness or paralysis on one side of the face.
 - Drooping of the mouth or eyelid.
 - Difficulty closing the eye, causing irritation or dryness.
 - Pain around the jaw or behind the ear.
 - Loss of taste and increased sensitivity to sound.
- **Duration and Recovery:** For about 80% of people, symptoms begin to improve within 3 weeks, with near-full recovery in 2-3 months. In some cases, recovery can take up to one year.
- **Long-term Effects:** While most make a full recovery, about 13% to 20% may experience lingering issues, such as permanent, mild weakness, or synkinesis (involuntary muscle movement, such as an eye closing when smiling).
- **Risk Factors:** It is more common in people who are pregnant (especially the third trimester), have recently had an upper respiratory infection (flu/cold), or have diabetes or obesity.
- **Treatment:** While there is no definitive cure, treatments to improve recovery speed include steroids and antiviral medicatio

Awake brain surgery, also called awake craniotomy, is a type of procedure performed on the brain while you are awake and alert. Awake brain surgery is used to treat some brain (neurological) conditions, including some brain tumors or epileptic seizures.

If your tumor or the area of your brain where your seizures occur (epileptic focus) is near the parts of your brain that control vision, movement or speech, you may need to be awake during surgery. Your surgeon may ask you questions and monitor the activity in your brain as you respond.

Risks

Some of the risks of awake brain surgery include:

- Changes in your vision
- Seizures
- Difficulty with speech or learning
- Loss of memory
- Impaired coordination and balance
- Stroke
- Swelling of the brain or too much fluid in the brain
- Meningitis
- Leaking spinal fluid
- Weak muscles

Brain Aneurysm

A brain (cerebral) aneurysm is a bulge in a weak area of a blood vessel in or around your brain. Most aneurysms are small and don't cause issues. But a ruptured brain aneurysm is life-threatening. The first sign of a ruptured brain aneurysm is usually a severe headache — the worst headache you've ever had. Seek medical care immediately if you have symptoms of a brain aneurysm rupture.

A brain aneurysm, also called a cerebral aneurysm, is a bulge in a weak area of an artery in or around your brain.

What is a brain aneurysm?

A brain aneurysm, also called a cerebral aneurysm, is a bulge in a weak area of an [artery](#) in or around your [brain](#). The constant pressure of blood flow pushes the weakened section outward, creating a blister-like bump.

When blood rushes into this bulge, the aneurysm stretches even farther. It's similar to how a balloon gets thinner and is more likely to pop as it fills with air.

Brain aneurysms can occur anywhere in your brain, but most of them form in the major arteries along the base of your skull. Approximately 10% to 30% of people who have a brain aneurysm have multiple aneurysms. The majority of brain aneurysms are small and don't cause symptoms.

An aneurysm can cause symptoms if it puts pressure on nearby nerves or brain tissue. If the aneurysm leaks or ruptures (bursts open), it causes bleeding in your brain. A ruptured brain aneurysm can be life-threatening and requires emergency medical treatment. As more time passes with a ruptured aneurysm, the likelihood of death or disability increases.

What happens when a brain aneurysm ruptures?

When it ruptures, blood spills ([hemorrhages](#)) into your surrounding brain tissue. The blood can put excess pressure on your brain tissue and make your brain swell. It usually causes a severe headache called a [thunderclap headache](#), in addition to other symptoms.

A ruptured brain aneurysm can cause serious health problems such as:

- **[Subarachnoid hemorrhage \(SAH\)](#):** Bleeding in the area between your brain and the thin tissues that cover and protect it (the arachnoid layer). About 90% of SAHs are due to ruptured brain aneurysms.
- **[Hemorrhagic stroke](#):** Bleeding in the space between your skull and brain.

This can result in permanent brain damage or other complications such as:

- **[Vasospasm](#):** This happens when blood vessels get narrower or clamp down and less oxygen reaches your brain.
- **[Hydrocephalus](#):** This happens when a buildup of cerebrospinal fluid or blood around your brain puts increased pressure on it.
- **[Seizures](#):** A seizure is a temporary, uncontrolled surge of electrical activity in your brain. It can make brain damage due to a ruptured aneurysm worse.
- **[Coma](#):** A state of prolonged unconsciousness. It can last days to weeks.
- **[Death](#):** Ruptured brain aneurysms result in death in about 50% of cases.

What is epilepsy surgery?

Epilepsy surgery is [brain surgery](#) to stop or reduce the number of [seizures](#) you have and/or their severity. Seizures are bursts of abnormal electrical activity between your brain's [nerve](#) cells. They can cause changes in your:

- Awareness.
- Behavior.
- Emotions.
- Muscle control.
- Sensations.

It's a big decision to have epilepsy surgery. It's important to note that undergoing surgery doesn't cure [epilepsy](#). There's no guarantee that you'll be seizure-free when you wake up from the procedure. But it may significantly improve your quality of life. Everyone's outcome is different.

Your healthcare provider will let you know if this is a safe option for you and what you can expect.

When do you need epilepsy surgery?

Surgery isn't the first option for epilepsy treatment. Your healthcare provider may consider surgery if:

- A non-epileptic condition causes seizures (for example, [brain tumor](#) or [arteriovenous malformation](#)).
- [Anti-seizure medication](#)(s) don't regulate seizures.
- The side effects of anti-seizure medications outweigh the benefits.
- Other treatment methods, like changing your eating habits ([ketogenic diet](#)), aren't effective.
- Seizures are frequent and making it difficult to function on a daily basis.

What are the types of epilepsy surgery?

There are many types of surgical procedures for epilepsy that vary based on the location of the seizures and/or the underlying cause. Common types of epilepsy surgery include:

- **Surgical resection:** Removing the part of your brain where the seizures start or removing the abnormal brain tissue that's causing the seizures. Examples include lesionectomy, [lobectomy](#), multilobar resection and [hemispherectomy](#).
- **Surgical disconnection:** Disconnecting parts of your brain or areas where nerve cells communicate to stop the spread of seizures. Examples include [corpus callosotomy](#) and multiple subpial transection.
- **Laser interstitial thermal therapy:** Using [MRI](#) scans to guide a small laser probe. This technique is used to destroy nerve cells at the seizure site. It's less invasive than other types of open surgery.
- **Neuromodulation (implanted devices):** Implanting a device and electrodes that send electrical signals to block or disrupt seizure activity at its source. Examples include [vagus nerve stimulation](#), [responsive neurostimulation](#) and [deep brain stimulation](#).
- **Implantation of electrodes:** Your provider may need to collect data by recording signals directly from your brain. They'll do this by carefully implanting electrodes into (or on the surface of) your brain. This information helps your surgeon decide upon a good surgical

strategy that's tailored toward the treatment of your seizures. Examples include stereoelectroencephalography and subdural grid implantation.

Surgery for Trigeminal neuralgia

Trigeminal neuralgia surgery aims to relieve severe facial pain when medication fails, primarily by addressing blood vessel compression on the nerve or damaging the nerve to block pain signals. Main options include [Microvascular Decompression](#) (MVD - most effective, long-term), [Gamma Knife Radiosurgery](#) (non-invasive), and [percutaneous rhizotomy](#) (needle-based, for elderly/frail).

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Key Surgical Procedures

- **Microvascular Decompression (MVD):** The preferred, curative surgery. A neurosurgeon performs a craniotomy (behind the ear) to place a sponge between the artery/vein and the trigeminal nerve, removing pressure.
- **Percutaneous Procedures (Needle Techniques):** Performed for patients who cannot tolerate major surgery, these involve inserting a needle through the cheek to damage the nerve and block pain.
 - [Radiofrequency Thermal Rhizotomy](#): Uses heat to destroy nerve fibers.
 - [Glycerol Injection](#): Injects a substance to damage the nerve.
 - [Balloon Compression](#): Inflates a small balloon to compress the nerve.
- **Stereotactic Radiosurgery (Gamma Knife):** A non-invasive, targeted radiation beam disrupts pain signals, often used when other surgeries are not suitable.

